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Subject : EDS

Screenshot Of Practical No.4 :

Practical Lab Assignment :

The image displays two screenshots of the CODETANTRA online IDE interface, showing a Python program for Pandas series creation and manipulation, and a program for DataFrame operations.

Top Screenshot: 4.1.1. Pandas - series creation and manipulation

The task description states: "Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations. Hint: You can group the Series based on whether each number is even or odd."

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

The code in the editor is as follows:

```
1 import pandas as pd
2
3 # Take inputs from the user to create a list of numbers
4 numbers = list(map(int, input().split()))
5
6
7 # Create a Pandas series from the list of numbers
8 s = pd.Series(numbers)
9
10 # Grouping by even and odd numbers and calculating the mean
11 # Grouping by even and odd numbers and calculating the mean
12 grouped = s.groupby(s % 2 == 0).mean()
13
14 # Display the mean of even and odd numbers with labels
15 grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
16 print("Mean of even and odd numbers:")
```

The test results show 3 out of 3 shown test case(s) passed and 3 out of 3 hidden test case(s) passed. The test case 1 output is as follows:

Expected output	Actual output
1 2 3 4 5 6 7 8 9 10	1 2 3 4 5 6 7 8 9 10
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd ---- 5.0	Odd ---- 5.0
Even ---- 6.0	Even ---- 6.0
dtype: float64	dtype: float64

Bottom Screenshot: 4.1.2. Dictionary to dataframe

The task description states: "A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation."

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the **Age** column.
- Display the DataFrame after deleting the column.

The code in the editor is as follows:

```
1 # Import pandas as pd
2
3 # Provided dictionary of lists
4 data = {
5     'Name': ['Alice', 'Bob', 'Charlie'],
6     'Age': [25, 30, 35],
7 }
8
9 # Convert the dictionary to a DataFrame
10 df = pd.DataFrame(data)
11
12 # Display the original DataFrame
13 print("Original DataFrame:")
14 print(df)
15
16 # Adding a new row
```

The test results show 1 out of 1 shown test case(s) passed and 1 out of 1 hidden test case(s) passed. The test case 1 output is as follows:

Expected output	Actual output
Original DataFrame:	Original DataFrame:
..... Name Age	 Name Age
0 Alice 25	0 Alice 25
1 Bob 30	1 Bob 30
2 Charlie 35	2 Charlie 35
New name: Susan	New name: Susan

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4.1.3. Student Information

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:
Refer to the displayed test cases for better understanding.

Sample Test Cases

studentin...pd

```
1 import pandas as pd
2
3 # Read the text file into a DataFrame
4 file = input()
5 data = pd.read_csv(file, sep='\t+', header=None, names=["Name", "Age", "Grade"])
6
7
8 # write your code here..
9 print("First five rows:")
10 print(data.head())
11
12 avg_age = round(data["Age"].mean(), 2)
13 print("Average age:", avg_age)
14
15 filtered_data = data[data["Grade"].isin(["A", "B"])]
16
17 print("Students with a grade of A or B")
```

Average time
0.021 s
21.00 ms

Maximum time
0.032 s
32.00 ms

1 out of 1 shown test case(s) passed
1 out of 1 hidden test case(s) passed

Test case 1

Expected output

studentdata.txt

First five rows:

--- Name Age Grade ---
0 John 25 A
1 Allin 22 B
2 Emma 24 A

Actual output

studentdata.txt

First five rows:

--- Name Age Grade ---
0 John 25 A
1 Allin 22 B
2 Emma 24 A

Terminal

Test cases

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Reset

Submit

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Lab Assignment :

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4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8 df['Date'] = pd.to_datetime(df['Date'])
9 df['Month'] = df['Date'].dt.strftime("%Y-%m")
10 df['Total Sales'] = df['Quantity'] * df['Price']
11
12 # Find the month with the highest total sales
13 sales_by_month = df.groupby('Month')['Total Sales'].sum()
14 best_month = sales_by_month.idxmax()
15 highest_sales = sales_by_month.max()
16
17 return f'Best month: {best_month} Total sales: {highest_sales}'
```

Average time: 0.015 s, Maximum time: 0.030 s, 14.67 ms, 30.00 ms

1 out of 1 shown test case(s) passed, 2 out of 2 hidden test case(s) passed

Test case 1: Expected output: sales_data.csv, Actual output: sales_data.csv, Best month: 2025-01, Total sales: \$1220.00

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4.2.2. Best Selling Product

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 product_sales = df.groupby('Product')['Quantity'].sum()
10 # Find the product with the highest total quantity sold
11 best_product = product_sales.idxmax()
12 highest_quantity = product_sales.max()
13
14 # Display the result
15 print(f'Best selling product: {best_product}')
16 print(f'Total quantity sold: {highest_quantity}')
17
```

Average time: 0.012 s, Maximum time: 0.023 s, 11.67 ms, 23.00 ms

1 out of 1 shown test case(s) passed, 2 out of 2 hidden test case(s) passed

Test case 1: Expected output: sales_data.csv, Actual output: sales_data.csv, Best selling product: Product A, Total quantity sold: 23

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4.2.2. Best Selling Product

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 product_sales = df.groupby('Product')['Quantity'].sum()
10 # Find the product with the highest total quantity sold
11 best_product = product_sales.idxmax()
12 highest_quantity = product_sales.max()
13
14 # Display the result
15 print(f'Best selling product: {best_product}')
16 print(f'Total quantity sold: {highest_quantity}')
17
```

Average time: 0.012 s, Maximum time: 0.023 s, 11.67 ms, 23.00 ms

1 out of 1 shown test case(s) passed, 2 out of 2 hidden test case(s) passed

Test case 1: Expected output: sales_data.csv, Actual output: sales_data.csv, Best selling product: Product A, Total quantity sold: 23

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4.2.3. City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by city and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 # write the code..
10 city_sales = df.groupby("city")["Quantity"].sum()
11 best_city = city_sales.idxmax()
12 # Display the result
13 print(f"City sold the most products: {best_city}")
14
```

Average time0.010 sMaximum time0.021 s1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1Expected outputActual output

sales_data.csvsales_data.csv

City sold the most products: Los AngelesCity sold the most products: Los Angeles

TerminalTest cases

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4.2.4. Most Frequently Sold Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pairs that was sold most frequently.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Explanation:

Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C
- 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product B
- 2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

- Product A and Product B: Appear in transactions on 2025-01-01 and 2025-01-03.
- Product A and Product C: Appear in transactions on 2025-01-02 and 2025-01-05.
- Product B and Product C: Appear in transactions on 2025-01-04.

Most Frequent Product Combinations:

- Product A and Product B (2 times)
- Product A and Product C (2 times)

Test Cases

frequentl...

```
1 import pandas as pd
2 from itertools import combinations
3 from collections import Counter
4
5 # Prompt user to input the file name
6 file_name = input()
7
8 # Read data from the specified CSV file
9 df = pd.read_csv(file_name)
10
11 # write the code
12 date_products = {}
13
14 for date, group in df.groupby("Date"):
15     products = group["Product"].unique()
16     if len(products) > 1:
17         date_products[date] = products
```

Average time0.011 sMaximum time0.024 s1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1Expected outputActual output

sales_data.csvsales_data.csv

Product A and Product B: 2 timesProduct A and Product B: 2 times

Product A and Product C: 2 timesProduct A and Product C: 2 times

TerminalTest cases

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4.2.5. Titanic Dataset Analysis and Data Cleaning

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

- Display the first 5 rows of the dataset.
- Display the last 5 rows of the dataset.
- Get the shape of the dataset (number of rows and columns).
- Get a summary of the dataset (using .info()).
- Get basic statistics (mean, standard deviation, etc.) of the dataset using .describe().
- Check for missing values and display the count of missing values for each column.
- Fill missing values in the 'Age' column with the median age.
- Fill missing values in the 'Embarked' column with the most frequent value (mode()).
- Drop the 'Cabin' column due to many missing values.
- Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below.

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171,725	5		S
2	1	1	Cunings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599,71.2833	C85	C	
3	1	1	McKinnen, Miss. Laina	female	26	0	0	STON/O2. 3101282,7.925	5		S
4	1	1	Tuttle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803,53.1	C123	S	
5	0	3	Allen, Mr. William Henry	male	35	0	0	37458,8.05			S
6	0	1	Moran, Mr. James	male	0	0	0	5381,0			

titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-dataset.csv')
6
7 import pandas as pd
8 import numpy as np
9
10 # Load the Titanic dataset
11 data = pd.read_csv('Titanic-dataset.csv')
12
13 # 1. Display the first 5 rows of the dataset
14 print(data.head())
15
16 # 2. Display the last 5 rows of the dataset
17 print(data.tail())
```

Average time0.092 sMaximum time0.092 s1 out of 1 shown test case(s) passed

Test case 1Expected outputActual output

PassengerId Survived Pclass Fare Cabin Embarked

0 0 3 7.2500 NaN S

1 1 1 71.2833 C85 S

2 1 1 7.9250 NaN S

3 1 3 53.1 NaN S

4 0 1 8.05 NaN S

TerminalTest cases

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4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
- Convert the 'Sex' column to numeric values (male: 0, female: 1).
- One-hot encode the 'Embarked' column, dropping the first category.
- Get the mean age of passengers.
- Get the median fare of passengers.
- Get the number of passengers by class.
- Get the number of passengers by gender.
- Get the number of passengers by survival status.
- Calculate the survival rate of passengers.
- Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,3,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,338877,0.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,S

1import pandas as pd
2import numpy as np
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6data['FamilySize'] = data['SibSp'] + data['Parch']
7
8import pandas as pd
9import numpy as np
10
11# Load the Titanic dataset
12data = pd.read_csv('Titanic-Dataset.csv')
13data['FamilySize'] = data['SibSp'] + data['Parch']
14# 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)
15data['IsAlone'] = np.where(data['FamilySize'] == 0, 1, 0)
16
17# 2. Convert 'Sex' to numeric (male: 0, female: 1)

Average time0.043 sMaximum time0.043 s1 out of 1 shown test case(s) passed
Test case 143 msDebug

Expected outputActual output
29,6991176470588229,69911764705882
14,454214,4542
3---4913---491
1---2561---256
2---1842---184
Name: Pclass, dtype: int64Name: Pclass, dtype: int64

TerminalTest cases

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4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Calculate the survival rate by class.
- Calculate the survival rate by embarkation location (Embarked_S).
- Calculate the survival rate by family size (FamilySize).
- Calculate the survival rate by being alone (IsAlone).
- Get the average fare by passenger class (Pclass).
- Get the average age by passenger class (Pclass).
- Get the average age by survival status (Survived).
- Get the average fare by survival status (Survived).
- Get the number of survivors by class (Pclass).
- Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,3,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,338877,0.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

1import pandas as pd
2import numpy as np
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6data['FamilySize'] = data['SibSp'] + data['Parch']
7data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
8data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10import pandas as pd
11import numpy as np
12
13# Load the Titanic dataset
14data = pd.read_csv('Titanic-Dataset.csv')
15data['FamilySize'] = data['SibSp'] + data['Parch']
16data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
17data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

Average time0.051 sMaximum time0.051 s1 out of 1 shown test case(s) passed
Test case 161 msDebug

Expected outputActual output
PclassPclass
1---0.6296301---0.629630
2---0.4728262---0.472826
3---0.2423633---0.242363
Name: Survived, dtype: float64Name: Survived, dtype: float64
Embarked_SEmbarked_S
+-----+-----

TerminalTest cases

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4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Get the number of survivors by gender (Sex).
- Get the number of non-survivors by gender (Sex).
- Get the number of survivors by embarkation location (Embarked_S).
- Get the number of non-survivors by embarkation location (Embarked_S).
- Calculate the percentage of children (Age < 18) who survived.
- Calculate the percentage of adults (Age >= 18) who survived.
- Get the median age of survivors.
- Get the median age of non-survivors.
- Get the median fare of survivors.
- Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,3,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,338877,0.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

1import pandas as pd
2import numpy as np
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
7
8import pandas as pd
9import numpy as np
10
11# Load the Titanic dataset
12data = pd.read_csv('Titanic-Dataset.csv')
13data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
14
15# 1. Get the number of survivors by gender
16survivors_by_gender = data[data['Survived'] == 1]['Sex'].value_counts()

Average time0.045 sMaximum time0.045 s1 out of 1 shown test case(s) passed
Test case 145 msDebug

Expected outputActual output
female---233female---233
male---109male---109
Name: Sex, dtype: int64Name: Sex, dtype: int64
male---468male---468
female---81female---81
Name: Sex, dtype: int64Name: Sex, dtype: int64

TerminalTest cases

